Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



SOLUTIONS ARCHITECT

Enterprise Integration | API & Data Architecture | AI-Enabled Solutions

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PROFESSIONAL SUMMARY

Key results: **98%** fewer reported ATC confusion events with production AI | **99.8%** availability across **200+ integration points and 86 systems** | new-integration lead time from **one week to one day** | **EUR 8M** cumulative savings

Solutions Architect experienced in airline IT, designing event-driven, API-first integration and data architecture for operations that cannot stop: the **high-transaction-volume integration backbone of a global carrier** on Apache Kafka, IBM MQ and REST/SOAP APIs, and an Airline Operational DataStore feeding **78 systems** across eleven downstream domains.

Moved a safety-critical problem from heuristics to production AI: a **fully AI-based callsign safety system** (Neo4j, Graph Attention Networks, PyTorch Geometric) for an **NP-hard** Quadratic Assignment Problem, alongside a real-time Early Warning System with zero restricted-airspace dispatch incidents since go-live.

Customer-side product owner and SME for NetLine, SchedConnect and flight planning across Turkish Airlines, AJet and SunExpress (a Lufthansa joint venture) through fleet growth from **60 to 500+ aircraft**: RFP/POC evaluations, technical due diligence and key-user training, with **54 partner carriers** in schedule data exchange and Lufthansa Systems, CAE and Jeppesen as vendors. Author of the integration standards and reference architectures adopted enterprise-wide.

RECENT PROFESSIONAL DEVELOPMENT

Career Break and Professional Development | **Istanbul** | **Sep 2024 – Present**

- Planned break for family and structured upskilling: **Harvard's CS50 AI** (12 Python projects: search, optimization, machine learning, neural networks, NLP), AWS Certified AI Practitioner and Cloud Practitioner (foundation-level AWS Well-Architected Framework), PSPO I and continued research on graph-based AI for aviation safety.

PROFESSIONAL EXPERIENCE

Principal Solutions Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

Official title: Distinguished Expert, the highest technical grade at Turkish Technology.

- Cut reported ATC confusion events by **98%**, from **50 to 1 per season** (sustained since), on up to ~1,500 daily peak-season flights, replacing the earlier heuristic with a fully AI-based callsign safety system aligned with EUROCONTROL CSST.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of roadmaps and technical direction for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget), sequencing change through **integration dependency management** and architectural constraints.
- Ran the NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of flight planning platforms (Jeppesen, CAE formerly FPM, Lido) through RFP/RFI, vendor management and business cases.
- Advised VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Held schedule mismatches at **zero** across two carriers and two reservations systems during AJet's migration from TROYA, the group's reservations and inventory platform, to a separate third-party PSS, running SchedConnect-based synchronization, inter-carrier schedule distribution and schedule change management in a transition architecture.
- Established design authority and design reviews bringing architecture decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, and supported regulatory audits and ICAO compliance.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) that delivered **9 systems to production**, as architecture and design decision owner, with Scrum Master support as required.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Shaped integration and technology strategy for next-generation aviation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety, IOCC and operational control systems.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and zero reported unplanned downtime on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Eliminated dispatches into restricted airspace (**18** in the 12 months before go-live, **zero** since) by engineering a real-time Early Warning System across NetLine and FPM: **72** automated checks and alarms on planned routes for up to **~1,500 daily** peak-season dispatch legs.
- Enabled **1.5M** disrupted passengers a year to re-accommodate online, even those en route to the aircraft, by extending PSS/DCS integration with the in-house PSS TROYA to irregular operations (IROPS/IRROPS) and disruption management, propagating IOCC decisions to inventory, reservations and departure control as they happened.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Engineered the reliability semantics behind the estate's **99.8%** availability: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery, dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Integrated a partner service behind an API gateway, handling API-key acquisition and token renewal on expiry.
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, automated testing strategies, and performance and security testing gating releases.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Lead Solutions Architect and Platform Owner | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Official title: Project Manager, Flight Operations.

- Designed and built **TKAODS** (Airline Operational DataStore) and evolved **OFPDS** (Operational Flight Plan DataStore, live since 2010), which together became the **single source of truth** for schedule and flight data.
- Took schedule and flight-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate flight-data interfaces** by routing every change through both platforms, 24/7.
- Architected both as **Tier-0** integration layers in front of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base) and flight planning.
- Owned enterprise-wide flight plan and flight data distribution for planning, operational and post-flight periods as the single integration point for **eleven main downstream domains**: scheduling, crew, load control, PSS/DCS interfaces, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** (peak-season average) for up to **~1,500 flights a day**, 24/7 and near-real-time, across summer and winter seasons.
- Consolidated, transformed and enriched every update through source-to-target mappings, ETL/ELT pipelines and integration rules under data quality and metadata controls.
- Architected batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** flight-data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to passenger-notification systems, triggering **75M notifications a year**, and flight data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Designed the enterprise operational data and integration architecture on the AODB (Airline Operational DataStore) pattern, standardizing flight, leg, schedule and operational flight-plan data airline-wide: logical data models, entities and relationships, canonical data structures, and the ESB pattern applied twice.

- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.
- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Drove technical direction, integration and platform modernization for airline operational systems with a **EUR 6.5M** total programme budget (scheduling, slot management, flight planning, IOCC) as lead architect with delivery accountability.
- Worked end to end: system and business analysis, software architecture and hands-on coding through release, operational handover and 24/7 production under ITIL.
- Managed NetLine's master and reference data (NetLine/Base) with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Built a reporting layer into both platforms on the same enriched data and fed it back into NetLine and the flight planning system, so enterprise reporting and operations shared one version of the data.
- Kept **850 NetLine users** working through the pandemic as NetLine product owner, coordinating the VMware solution.

Integration Solution Design Lead | Turkish Airlines | Istanbul, Turkiye | Jan 2007 – Jun 2014

Official title: Acting Team Leader.

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency overflight cost engine built on a data pipeline of country entry/exit timestamps and segment distances.
- Architected the schedule-to-PSS (TROYA) message integration under IATA SSIM Chapter 4 (SSM) and Chapter 5 (ASM) over Type B messaging, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated flights and disruption reason codes.
- Introduced SchedConnect in 2007 for inter-carrier schedule enrichment and distribution, establishing recurring schedule data exchange with **54 partner carriers** under IATA data standards.
- Headed solution design and delivery for flight operations, crew and flight-planning integrations across global aviation vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Built **OFPDS** in 2010 to cut mainframe dependency: flight plans from the new planning system landed in Oracle and the data warehouse, easing SAP integration; the fuel efficiency team rated them far more accurate than the earlier Jeppesen feed.
- Developed the first heuristic Callsign Similarity & Assignment System (phonetic, visual and semantic similarity), the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | Turkish Airlines | Istanbul, Turkiye | Apr 2000 – Jan 2007

Official title: Analyst Programmer.

- Contributed to the first production implementation of Lufthansa Systems NetLine (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base), the platform that scaled with fleet growth from **60 to 500+ aircraft** and from 10M to 85M passengers.
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch: worldwide airline schedules and partner codeshare flights in, a PDF timetable for web download and print out, so the scheduling department began preparing the timetable books itself (Java, C, PL/SQL).
- Administered NetLine at application level with 24/7 availability responsibility (upgrades and patches on AIX, database schema upgrades) and built its bi-directional real-time interfaces to the IBM mainframe on IBM MQ.
- Drove the technical work of the first major legacy modernization (re-platforming) from day one as an implementation team member, moving schedule and operations control off the IBM mainframe (ADABAS/Natural) onto Oracle on Unix (AIX) and NetLine, retiring mainframe-era technical debt.
- Designed and built **Flight Watch**, a 24/7 world map tracing every Turkish Airlines flight in the air with simulated real-time positions: architecture decisions, mainframe integration over IBM MQ and UDP multicast (Java, Oracle, ILOG JViews Maps).

CORE COMPETENCIES & TECHNICAL SKILLS (hands-on)

Integration and API Governance: Event-Driven Architecture, API Architecture, API lifecycle management, API versioning and backward compatibility, integration standards and governance, Apache Kafka (schema registry), IBM MQ, ESB, REST/SOAP APIs; GraphQL (design level).

Data Architecture and Governance: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, master data management (MDM), data lineage through source-to-target mappings, data quality and metadata, schema evolution.

Aviation Operations and Airline IT: Integrated Operations Control Center (IOCC), operational and tactical flight planning (Jeppesen; FPM, now CAE Flight Planning Solutions; Lido), schedule management, ACARS air-ground messaging over the SITA network; DWH, SAP and MRP interfaces.

IATA Schedule Data Standards: SSIM Chapter 7 (Presentation and Transfer of a Schedule Data Set); schedule-to-PSS distribution across three group carriers.

Analytics & Graph / Advanced Analytics and AI: Neo4j and graph databases, PyTorch and PyTorch Geometric, MLOps, statistical analysis, data quality and metadata management.

Engineering — Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML, object-oriented design and system design; multithreaded real-time and database-driven systems; 50+ applications and integrations, with incident analysis and performance tuning.

Executive — Technical Leadership and Delivery: Migration and transition architecture, transformation governance, executive (C-level) stakeholder management, customer consulting and workshop facilitation.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• Project Management Professional (PMP), Credential ID 1225730	PMI	2008
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	2013
• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• CS50's Introduction to Artificial Intelligence with Python (12 projects)	Harvard University (CS50)	2026
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• AI, Cloud and Container Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI); Docker Foundations Professional Certificate	Oracle, IBM, PMI, Docker	2026
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	2007